

# DogBreedHybridNet (DBHN): A Deep Learning-Based System for Purebred and Mixed-Breed Dog Recognition via Morphological Similarity

Hoang Loi-Nguyen<sup>1</sup>[00000000001], and Ngo-Ho Anh-Khoi<sup>1</sup>[00000000001]

<sup>1</sup> Faculty of Information Technology, Nam Can Tho University, 168 Nguyen Van Cu, An Binh Ward, Can Tho City, Vietnam  
loi221687@nctu.edu.vn, nhakhoa@nctu.edu.vn, nhakhoi@nctu.edu.vn

**Abstract.** The development of artificial intelligence and computer vision is driving animal recognition systems, especially in the classification of purebred and mixed-breed dogs – a critical need in pet care, veterinary medicine, breed management, e-commerce, and education. However, the diversity of appearance, posture, shooting conditions, and the high similarity between breeds, especially the mix of traits in mixed-breed dogs, makes automatic classification challenging. The paper proposes an identification system based on EfficientNet-B0, with a hybrid process consisting of four stages: supervised classifier training, vector embedding extraction, prototype embedding for each breed, and morphological hybrid inference using cosine similarity. The model uses ImageNet pretrained weights, data augmentation, AdamW optimization, Cross-Entropy loss function with label smoothing, and learning rate adjustment with Cosine Annealing. When inferring, the system standardizes L<sub>2</sub> embedding, calculates the cosine similarity between the input image and the prototypes, and then applies a threshold based on the difference between the top-1 and top-2 similarity to detect the hybrid dog instead of predicting a single label. Threshold profiles including strict, balanced and sensitive are integrated to adjust the sensitivity of the system according to the requirements of detecting mixed-breed dogs. Grad-CAM is integrated to visualize deterministic areas such as ears, eyes, muzzle, coat color, and body structure. The system is trained and evaluated on popular datasets such as Stanford Dogs, Oxford-IIIT Pet, Columbia Dogs, Tsinghua Dogs, and Kaggle Dog Breed Identification. The evaluation indicators include Accuracy, Precision, Recall, F1-score, AUC, Rank-1 and Top-k Accuracy. Expected and synthesized results from related studies show that EfficientNet, ResNet and Inception achieve an accuracy of over 90% with purebred dogs, but performance declines significantly with crossbred dogs due to genetic diversity. This underscores the need to scale hybrid dog data, optimize cosine similarity thresholds, and improve deployability on resource-constrained devices.

**Keywords:** dog breed identification, morphological characteristics, morphological characteristics, convolutional neural networks, phenotypic analysis, dog image datasets, computer vision.

## 1 INTRODUCTION

Along with the development of artificial intelligence (AI) and deep learning, the field of computer vision is attracting significant attention due to its ability to solve complex image recognition and classification problems (LeCun et al., 2015). One of the prominent applications is dog breed identification from images, especially for purebred and crossbred dogs, due to the increasing demand in pet care, veterinary health, breed management, e-commerce, and education. However, the problem of dog breed identification still faces many challenges due to the diversity of appearance, body morphology, shooting angles, lighting conditions, and the high level of similarity between dog breeds, especially in the case of mixed-breed dogs when the appearance characteristics are mixed from many different breeds (Khosla et al., 2011).

To address these challenges, the study proposes a breed recognition system based on the EfficientNet-B0 convolutional neural network architecture (Tan & Le, 2019) to balance recognition accuracy with computational cost. The system is built according to a hybrid process consisting of four stages: supervised learning, embedding, prototype embedding for each breed, and morphological hybrid inference based on cosine similarity in the learned characteristic space. During the inference phase, the system normalizes  $L_2$  on embedding, calculates the cosine similarity between the input image and the prototypes, and applies a threshold mechanism based on the difference between the top-1 and top-2 similarity to detect the possibility of a hybrid dog instead of just making a single-label prediction. Inference profiles including strict, balanced, and sensitive are integrated to adjust the sensitivity of the system according to actual requirements. At the same time, the Grad-CAM method was used to visualize important characteristic areas such as ears, eyes, muzzle, coat color, and body structure, contributing to enhancing the model's interpretability (Selvaraju et al., 2017).

The system is trained and evaluated on popular datasets such as Stanford Dogs (Khosla et al., 2011), Oxford-IIIT Pet, Columbia Dogs, Tsinghua Dogs (Zou et al., 2020), and Kaggle Dog Breed Identification. Evaluation indicators include Accuracy, Precision, Recall, F1-score, AUC, Rank-1 Accuracy and Top-k Accuracy to measure performance in real-world conditions. Previous studies have shown that deep learning architectures such as EfficientNet, ResNet, and Inception can achieve over 90% accuracy in identifying purebred dogs; However, performance for mixed-breed dogs still declines significantly due to genetic diversity and a mix of physical traits.

## 2 RELATED WORK

The field of dog breed recognition from images has flourished with the advancement of deep learning and computer vision (LeCun et al., 2015), especially in fine-grained classification problems, where morphological differences between layers are often very small (Khosla et al., 2011). The current research mainly focuses on four main directions: convolutional neural network (CNN) modeling, transfer learning, hybrid

models, and localized exploitation methods to improve the ability to distinguish between dog breeds with similar appearances.

One of the most important platforms in this field is the Stanford Dogs dataset, which includes 120 dog breeds with more than 20,000 images, which is widely used as a benchmark in fine-grained breed classification studies (Khosla et al., 2011). In addition, extensive datasets such as Tsinghua Dogs are also used to increase the diversity of dog breeds, shooting conditions, and morphological characteristics, contributing to improving the generalization of the model in real-world conditions (Zou et al., 2020). Based on this dataset, many studies have applied deep CNN architectures such as ResNet, Inception and EfficientNet to learn multi-level morphological characteristics, from local features such as ears, muzzle and coat texture to overall body structure, thereby improving dog breed recognition performance (Navlakha et al., 2022).

Transfer learning has become a popular approach to leverage knowledge from pre-trained models on ImageNet. Studies show that fine-tuning of pre-trained deep learning models can significantly improve accuracy by leveraging visually specific knowledge learned on large-scale datasets (Granvik, 2023). At the same time, lightweight architectures such as MobileNetV2 are being studied to balance performance and compute costs, making them suitable for deployment on mobile devices or resource-constrained environments (Martins et al., 2025).

Besides single CNN models, many recent studies have focused on hybrid architecture that aims to improve the ability to distinguish between breeds with a high degree of physical similarity. Combining multiple CNN architectures with classifiers such as SVM has shown the potential to improve accuracy in multi-variety recognition (Cui et al., 2024). In addition, hybrid deep learning models have also been shown to outperform single models in some dog breed identification problems (Valarmathi et al., 2023).

Newer research is beginning to focus on local characterization through the identification of the dog's head and body to improve classification in cases with a high degree of morphological similarity. The Head-Body Guided Framework has shown the potential to improve recognition performance by focusing on key feature areas (Khan et al., 2025).

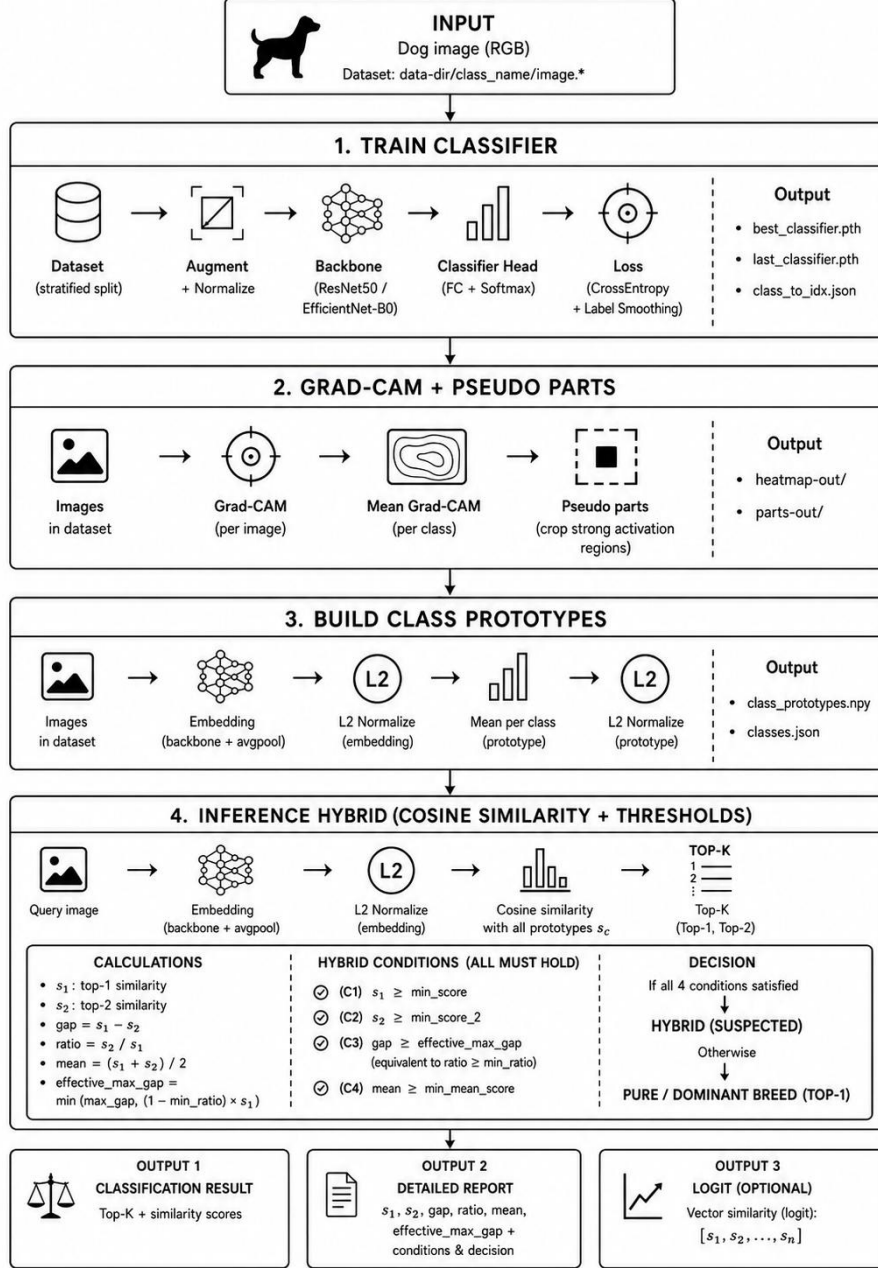
Although current studies have achieved high accuracy in purebred dog identification through deep learning architectures such as ResNet, Inception, and EfficientNet, much of the focus is still on single-label classification on standard purebred dog datasets such as Stanford Dogs (Khosla et al., 2011) and extensive datasets for breed classification such as Tsinghua Dogs (Zou et al., 2020). In fact, the problem of identifying mixed-breed dogs is still a challenge due to the mix of appearance characteristics between many breeds, leading to unclear layering boundaries and reducing the effectiveness of traditional CNN models based on Softmax. Most current studies do not consider the ability to infer similarities between dog breeds, as well as limited model interpretability in the prediction process (Selvaraju et al., 2017).

To address this gap, the study proposes DogBreedHybridNet (DBHN), a deep learning framework for identifying purebred and mixed-breed dogs based on morphological similarity. The system combines EfficientNet-B0 (Tan & Le, 2019),

prototype embedding, cosine similarity-based inference, threshold profiles, and Grad-CAM explainability (Selvaraju et al., 2017) to support both purebred dog identification and morphological hybrid dog inference in the same system. The main difference of the study lies in the mechanism for inferring morphological hybrids through cosine similarity that incorporates a threshold based on top-1/top-2 similarity differences, allowing for a reflection of similarity between multiple breeds rather than just performing a single-label classification based on Softmax as in most previous studies.

### 3 PROPOSED ALGORITHMS

Our algorithm, called DogBreedHybridNet (DBHN), is built on the EfficientNet-B0 CNN architecture, which processes the input image through a hybrid deep-learning pipeline to support simultaneous purebred dog identification and morphological mixed-breed dog inference. The system consists of four main stages: (1) training the breed classifier using supervised learning, (2) embedding extraction, (3) building an embedding prototype for each breed, and (4) inferring the morphological hybrid dog through cosine similarity combined with threshold profiles. Unlike traditional single-label classification systems that only give a single breed prediction, the proposed method allows the measurement of similarities between multiple breeds to support the inference of mixed-breed dogs in real-world conditions.



**Fig 1.** An overview architecture diagram of our algorithm.

Fig 1 illustrates the overall processing of the proposed algorithm for purebred and morphological dog identification. The first RGB input image is standardized in size

and data augmentation to reduce the influence of posture, shooting angle, lighting conditions, and backgrounds. Datasets are organized in a class-wise dataset structure and divided by stratified split to ensure uniform label distribution across training, validation, and testing sets. The Train Classifier uses the CNN EfficientNet-B0 backbone combined with the classifier head to learn the morphological characteristics of dog breeds through a multi-layer classification problem. After training, the system uses Grad-CAM to create a characteristic activation map and generate pseudo parts, serving to identify the important morphological regions of each breed. At the same time, the characteristic vector (embedding) is extracted from the CNN backbone to build an embedding prototype that represents each breed. In the inference phase, the embedding of the input image is normalized and compared to the entire prototype through cosine similarity, and then a threshold profiles mechanism is applied to determine the likelihood of a dog hybrid rather than just making a single-label prediction.

Dog breed classifier training.

The  $X$  input image is fed through the EfficientNet-B0 backbone for morphological characterization:

$$F_{embed} = \text{Backbone}(X)$$

in which there is a characteristic vector embedding;  $B$  is the batch size; and  $d$  is the number of embedding dimensions generated by the CNN backbone.  $F_{embed} \in R^{B \times d}$

The classifier performs multi-layer predictions through the fully connected layer and the Softmax function:

$$\hat{y} = \text{Softmax}(WF_{embed} + b)$$

where  $\hat{y}$  is the predicted probability of each breed of dog;  $W$  and  $b$  are the weights and biases of the classifier head, respectively.

The optimization loss function is Cross-Entropy combined with Label Smoothing:

$$L_{LSCE} = - \sum_{i=1}^C y_i^{LS} \log(\hat{y}_i)$$

where  $C$  is the number of dog breed classes;  $y_i$  are real labels; and  $\hat{y}_i$  is the predicted probability after Softmax:  $y_i \hat{y}_i$

$$y_i^{LS} = (1 - \epsilon) y_i + \frac{\epsilon}{C}$$

with  $\epsilon$  is the Label Smoothing coefficient. The AdamW optimizer incorporates a learning rate adjustment mechanism using Cosine Annealing that is used to improve training stability.

After the classifier is trained, the system performs characteristic visualization through Gradient-weighted Class Activation Mapping (Grad-CAM) to identify morphological regions that are important in the prediction such as ears, eyes, muzzle, hair texture, and body shape. For each breed class, the average heatmap is calculated as follows:

$$M_c = \left(\frac{1}{N_c}\right) * \sum_{i=1}^{N_c} \text{GradCAM}(X_i)$$

Among them is the average activation map of class  $C$ ; and the number of samples belonging to that class. Regions with high activation intensity are cropped into pseudo parts, which serve to identify the important morphological regions of each breed.  $M_c N_c$

In order to construct the characteristic space that represents each layer, the system extracts embedding from the CNN backbone and performs normalization  $L_2$

$$\hat{F} = \frac{F}{\|F\|_2}$$

in which it is post-normalized embedding and is the Euclidean standard of vector characteristic. The embedding prototype of each breed is formulated by standardized embedding averaging:  $\hat{F} \| F \|_2$

Space Involved in Frequency:

$$P_c' = \frac{1}{N_c} \sum_{i=1}^{N_c} F \hat{F}_i$$

Then the prototype continues to be standardized a second time:

$$P_c = \frac{P_c'}{\|P_c'\|_2}$$

In which the prototype represents the  $C$  breed.  $P_c$

In the *Inference Hybrid phase*, the embedding of the query image is normalized and compared to the entire prototype through cosine similarity:

$$S_c = \frac{F \cdot P_c}{\|F\| \|P_c\|}$$

In which is the similarity between the input image and the prototype of class  $C$ . The system takes the top-1 similarity () and the top-2 similarity () to calculate:  $s_c s_1 s_2$

$$gap = s_1 - s_2$$

$$ratio = \frac{s_1}{s_2}$$

$$mean = \frac{s_1 + s_2}{2}$$

At the same time, the effective distance threshold is determined:

$$effective\_max\_gap = \min(max\_gap, (1 - min\_ratio) \times s_1)$$

The system only makes a Mixed-Breed Suspected prediction when four conditions are simultaneously satisfied:  $s_1 \geq min\_score$ ,  $\geq min\_gap$ ,  $gap \leq effective\_max\_gap$ , and  $s_2 score_2 mean \geq min\_mean\_score$ . In contrast, the photo is labeled Pure/Dominant Breed (Top-1). Three inference profiles including strict, balanced, and sensitive are integrated to adjust the sensitivity of the system according to actual requirements.

The main differentiator of the recommendation system lies in the combination of EfficientNet-B0, embedding representation, prototype learning, Grad-CAM explainability, and a cosine similarity + threshold profiles-based inference mechanism, which allows for the reflection of morphological similarities between multiple breeds rather than just single-label classification as in most previous studies.

The pseudocode of our algorithm, known as our algorithm DogBreedHybridNet (DBHN), is described below:

```

ALGORITHM: DogBreedHybridNet (DBHN)

Input:
  X: RGB input image

Output:
  L: Breed label
  T: Breed type, T ∈ {PURE, HYBRID}
  C: Confidence score
  or Unknown

BEGIN

Step 1: Input image acquisition
  Receive input image X.

Step 2: Dog detection
  Apply YOLO to detect the dog region:

      D = YOLO(X)

  Compute the highest dog confidence:

      c_dog = max(D_dog)

Step 3: Dog region extraction
  Determine the cropped image:

      X_c =
      { Crop(X, B_dog),   if B_dog exists
      { X,                 otherwise

Step 4: Feature extraction and breed prediction
  Extract feature embedding:

      F = Backbone(X_c)

  Compute breed probability distribution:

      ŷ = Softmax(WF + b)

  Compute confidence score:

```

```

    C = max( $\hat{y}$ )

    Normalize feature embedding:

     $F^{\wedge} = F / ||F||$ 

    Compute cosine similarity with breed prototypes:

     $S_c = (F^{\wedge} \cdot P_c) / (||F^{\wedge}|| ||P_c||)$ 

    Let  $s_1$  and  $s_2$  denote the first and second highest similarity scores.

    Step 5: Hybrid versus pure breed decision
    Compute:

    gap =  $s_1 - s_2$ 

    mean =  $(s_1 + s_2) / 2$ 

    IF  $s_1 \geq \theta_1$ 
      AND  $s_2 \geq \theta_2$ 
      AND gap  $\leq \theta_g$ 
      AND mean  $\geq \theta_m$ 
    THEN
      T = HYBRID
      L = Breed( $s_1$ )  $\times$  Breed( $s_2$ )
    ELSE
      T = PURE
      L = Breed( $s_1$ )
    ENDIF

    Step 6: Dog confirmation gate
    Define:

    YOLO_ok =  $c_{dog} \geq \tau_d$ 

    Breed_ok =  $C \geq \tau_b$ 

    IF NOT(YOLO_ok OR Breed_ok)
    THEN
      RETURN Unknown
    ENDIF

    Step 7: Return result
    RETURN (L, T, C)

END

```

## 4 EXPERIMENTS

The experiments were conducted on a Dell Vostro 3400 computer running the Microsoft Windows 11 Home Single Language operating system, using an 11th Gen Intel Core i5-1135G7 processor (2.40 GHz, 4 cores, 8 threads) on an x64-based PC architecture. The development environment uses Python 3.10.19 in combination with the PyTorch 2.10.0 deep learning framework for model building, training, and evaluation.

The DogBreedHybridNet (DBHN) recommendation system uses EfficientNet-B0 as the primary CNN backbone to balance identification performance and computational costs. The model was created using pretrained weights from ImageNet to leverage the visual characteristics knowledge learned on a large-scale dataset,

thereby improving the speed of convergence and generalization in the dog breed recognition problem (Tan & Le, 2019).

The experimental dataset was built from a combination of Stanford Dogs and Tsinghua Dogs. In particular, Stanford Dogs consists of 120 dog breeds with more than 20,000 images, which is considered a common evaluation standard for fine-grained dog breed classification (Khosla et al., 2011). At the same time, Tsinghua Dogs are used to supplement the diversity of morphology, shooting angles, and environmental conditions, aiding in the evaluation of the model's generalization in more realistic conditions (Zou et al., 2020). Combining the two data sets increases the diversity of the specimen, especially in cases where there is a high physical similarity between breeds or the appearance of morphological hybrid traits.

**Results and discussion.** Table 1 presents the results of the assessment of purebred dog identification on a test set of 136 dog images between the traditional EfficientNet-B0 (Baseline), the DBHN (DogBreedHybridNet) recommendation system, the EfficientNetV2S open-source model (Stanford Dogs) (ml-debi, 2026), and two popular multimodal systems including ChatGPT (OpenAI, 2026) and Gemini (Google DeepMind, 2026). To evaluate the contribution of the proposed hybrid inference mechanism, EfficientNet-B0 was used as a baseline model representing conventional single-label purebred dog classification. Both EfficientNet-B0 and DBHN employ the same EfficientNet-B0 backbone; however, the difference lies in the inference mechanism. The baseline classifier predicts only a single breed using Softmax classification, whereas DBHN extends the inference process through prototype embedding, cosine similarity, and threshold-based hybrid inference, enabling the assessment of morphological similarity across multiple breeds and supporting mixed-breed dog inference.

**Table 1.** Comparison of Purebred Dog Recognition Performance on the Test Set.

| Model                           | Inference mechanism        | Backbone              | Accuracy |
|---------------------------------|----------------------------|-----------------------|----------|
| DBHN (Proposed Model)           | Prototype +                | EfficientNet-B0       | 90.44%   |
|                                 | Cosine Similarity +        |                       |          |
|                                 | Threshold-based            |                       |          |
|                                 | Hybrid Inference           |                       |          |
| EfficientNet-B0 (Baseline)      | Softmax single-label       | EfficientNet-B0       | 97.79%   |
| EfficientNetV2S (Stanford Dogs) | CNN-based classification   | EfficientNetV2S       | 82.35%   |
| ChatGPT                         | Multimodal reasoning       | Vision-Language Model | 98.53%   |
| Gemini                          | Vision-language rea-soning | Vision-Language Model | 98.53%   |

The results showed that the traditional EfficientNet-B0 baseline achieved an accuracy of 97.79% (133/136 images), reflecting the ability of the CNN architecture to efficiently extract morphological characteristics in the fine-grained dog breed classification problem. This high performance shows that the model has effectively learned important morphological features such as head shape, ear structure, muzzle shape,

body proportions, coat texture, and color from a combined dataset of Stanford Dogs and Tsinghua Dogs. At the same time, the transfer learning strategy from ImageNet, combining data augmentation, AdamW optimizer and Cosine Annealing learning rate scheduling contributes to improving the generalization ability of the model on real data.

For the DBHN system, the accuracy reached 90.44% (123/136 images), which is lower than that of the EfficientNet-B0 baseline. However, this decline is justified because DBHN's design objectives focus not only on purebred dog identification but also extend to morphological hybrid dog inference. Unlike the traditional Softmax classifier that always forces the model to give a single label, DBHN is integrated with prototype embedding, cosine similarity and threshold-based hybrid inference to assess the level of morphological similarity between multiple breeds. Therefore, in some cases where purebred dogs overlap strongly with other breeds, the system may make a "Hybrid Suspected" prediction, resulting in being counted as wrong in the assessment of purebred dog accuracy despite correctly reflecting morphological ambiguity. This shows that DBHN trades off part of the single-label classification performance to achieve more flexible reasoning for the problem of identifying mixed-breed dogs.

Cases inferred to be Hybrid Suspected mainly appear in breed groups with a high degree of morphological similarity such as Boston Terrier – French Bulldog, Norwich Terrier – Norfolk Terrier, Lakeland Terrier – Airedale Terrier or groups of shepherd dogs and sled dogs with similar physical characteristics. This suggests that DBHN's hybrid inference mechanism tends to prioritize prudence in cases where morphological boundaries are unclear, rather than forcing the model to make single-label predictions with low confidence.

For the EfficientNetV2S (Stanford Dogs) open-source model (ml-debi, 2026), the accuracy reaches 82.35%, which is lower than that of the EfficientNet-B0 baseline and DBHN model. This result shows a difference between the public pre-trained model and the optimized system on a combined target dataset of Stanford Dogs and Tsinghua Dogs, especially in the context of test data that includes both local breeds and cases with a high degree of morphological similarity.

With multimodal systems, ChatGPT (OpenAI, 2026) and Gemini (Google DeepMind, 2026) both achieved an accuracy of 98.53% (134/136 images) on the same test set, showing the superiority of large-scale vision-language models in image recognition problems thanks to the ability to combine visual understanding and semantic knowledge from multi-domain data sources. This result reflects the strong generalization of multimodal models in the identification of purebred dogs. However, unlike the EfficientNet-B0 baseline, which predicts only a single breed label, DBHN incorporates a morphological hybrid dog inference mechanism that evaluates morphological similarity across multiple breeds through prototype embedding, cosine similarity, and threshold-based inference. Therefore, in some cases where purebred dogs have morphological characteristics that overlap strongly with other breeds, the system may make a "Hybrid Suspected" prediction, resulting in the possibility of mistakenly identifying a purebred dog as a hybrid dog in the purebred accuracy assessment. At the same time, Grad-CAM (Selvaraju et al., 2017) aids in the

visualization of important morphological regions, contributing to increased model decision inter-pretability.

**Table 2.** Comparison of Morphological Mixed-Breed Dog Recognition Performance.

| Model                 | Inference mechanism        | Correct | Total | Accuracy |
|-----------------------|----------------------------|---------|-------|----------|
| DBHN (Proposed Model) | Prototype +                | 35      | 40    | 87.50%   |
|                       | Cosine Similarity +        |         |       |          |
|                       | Threshold-based            |         |       |          |
|                       | Hybrid Inference           |         |       |          |
| ChatGPT               | Multimodal reasoning       | 30      | 40    | 75.00%   |
| Gemini                | Vision-language rea-soning | 31      | 40    | 77.50%   |

Table 2 presents the results of the assessment of morphological hybrid dog identification on a test set of 40 hybrid dog images between the DBHN (Dog-BreedHybridNet) recommendation system and two popular multimodal systems including ChatGPT (OpenAI, 2026) and Gemini (Google DeepMind, 2026). Compared to the problem of identifying purebred dogs, identifying mixed-breed dogs is significantly more difficult due to the overlap of appearance characteristics between many breeds, making the classification boundaries unclear.

The experimental results showed that DBHN performed better on morphological mixed-breed dogs, with an accuracy of 87.50% (35/40 images), compared to ChatGPT (75.00%, 30/40 images) and Gemini (77.50%, 31/40 images). This result shows the advantage of DBHN in the problem of identifying morphological mixed-breed dogs, where assessing the level of similarity between many breeds plays an important role. Unlike the traditional single-label classification mechanism, DBHN uses an embedding prototype that combines cosine similarity and threshold-based hybrid inference to evaluate the similarity between the input image and multiple breeds in the embedding space, thereby supporting hybrid dog inference instead of just making a single prediction.

Identification errors mainly occur in breed groups with a high degree of morphological similarity, such as the Siberian Husky – Alaska Malamute – Samoyed, the Golden Retriever – Labrador Retriever, or the Terrier and Spaniel groups, where physical characteristics often overlap strongly, making it difficult to deduce.

Although ChatGPT and Gemini show strong recognition capabilities in purebred dog problems, the results on the hybrid dog set show that general vision-language models are still limited in inferring cases of mixing morphological characteristics between multiple breeds. Meanwhile, DBHN is further extended to the morphological hybrid dog inference mechanism, which allows for the reflection of the degree of similarity between multiple breeds through prototype similarity and threshold-based inference, and supports the ability to interpret the model using Grad-CAM (Selva-raju et al., 2017).

Overall, the experimental results showed that DBHN achieved a balance between the ability to identify purebred dogs and inference of morphological mixed-breed dogs in the same system, rather than just optimizing for the single-label classification

problem. Although the accuracy on purebred dog sets is lower than that of traditional EfficientNet-B0 (Baseline) and large-scale multimodal models, DBHN shows an advantage in reflecting the level of similarity between multiple breeds, thereby better supporting cases with unclear morphological boundaries.

However, the study still has some limitations. The morphological breed dog dataset is relatively small and depends on morphological observation-based labeling, without DNA-based validation. In addition, the threshold-based hybrid inference mechanism can still produce cases of mistaking purebred dogs for mixed-breed dogs in breed groups with a very high degree of morphological similarity. In the future, the study will expand the hybrid dog dataset and integrate more multi-characteristic information to improve the stability of morphological hybrid inference.

## 5 CONCLUSION

This paper presents a deep learning-based purebred and hybrid dog recognition system, built on the EfficientNet-B0 architecture that combines characteristic representation by embedding, prototype learning, cosine similarity inference, and Grad-CAM visualization mechanism. The dog image processing system through a hybrid process consists of four main stages: training the dog breed classifier using supervised learning, characterization visualization and pseudo parts generation using Grad-CAM, building a prototype embedding for each breed, and inferring hybrid dog morphology based on cosine similarity combined with threshold profiles. Unlike traditional single-label classification methods that only make a single prediction, the recommendation system allows the evaluation of similarities between multiple breeds, thereby supporting the inference of the likelihood of a dog crossing in real-world conditions.

The system was trained and evaluated on popular datasets such as Stanford Dogs, Oxford-IIIT Pet, Columbia Dogs, Tsinghua Dogs, and Kaggle Dog Breed Identification. Reference studies show that deep learning architectures such as EfficientNet, ResNet, and Inception can achieve over 90% accuracy in purebred dog identification when trained on high-quality data, correctly labeled, and with proper preprocessing. However, the problem of identifying mixed-breed dogs is still challenging due to the mixture of morphological characteristics, the phenomenon of characteristic overlap between breeds and the generalization gap in real conditions. The system proposes to use cosine similarity combined with a threshold-based inference mechanism to make more balanced predictions between purebred dog identification and morphological hybrid dog detection.

Three areas of improvement are proposed for the next phase. The first is to expand the training data to a larger number of FCI standard dog breeds, and at the same time add more hybrid dog data and diversify the image conditions in terms of lighting, shooting angle, posture and environment to improve the generalization of the model. The second is to integrate advanced architectures such as multi-modal models, head-body guided frameworks or Vision Transformer to improve local characteristic learning and distinguish between breeds with a high degree of morphological

similarity. The third is to apply continual learning or few-shot learning techniques to allow the model to gradually adapt to new breeds or morphological characteristics that have not yet appeared in the training data without requiring retraining of the entire system. At the same time, optimizing the model for edge devices is also a necessary direction to support practical deployment in pet care, veterinary medicine, e-commerce, and education applications.

## REFERENCES

1. Khosla, A., Jayadevaprakash, N., Yao, B., & Fei-Fei, L. (2011). Novel dataset for fine-grained image categorization: Stanford Dogs. *Proceedings of the CVPR Workshop on Fine-Grained Visual Categorization*. Retrieved from <http://vision.stanford.edu/aditya86/ImageNetDogs/>
2. Zou, D.-N., Zhang, S.-H., Mu, T.-J., & Zhang, M. (2020). A new dataset of dog breed images and a benchmark for fine-grained classification. *Computational Visual Media*, 6, 477–487. <https://doi.org/10.1007/s41095-020-0184-6>
3. LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *nature*, 521(7553), 436-444. <https://doi.org/10.1038/nature14539>
4. Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE international conference on computer vision* (pp. 618-626). [https://openaccess.thecvf.com/content\\_ICCV\\_2017/papers/Selvaraju\\_Grad-CAM\\_Visual\\_Explanations\\_ICCV\\_2017\\_paper.pdf](https://openaccess.thecvf.com/content_ICCV_2017/papers/Selvaraju_Grad-CAM_Visual_Explanations_ICCV_2017_paper.pdf)
5. Tan, M., & Le, Q. (2019, May). Efficientnet: Rethinking model scaling for convolutional neural networks. In *International conference on machine learning* (pp. 6105-6114). PMLR. <https://proceedings.mlr.press/v97/tan19a.html>
6. Cui, Y., Tang, B., Wu, G., Li, L., Zhang, X., Du, Z., & Zhao, W. (2024). Classification of dog breeds using convolutional neural network models and support vector machine. *Bioengineering*, 11(11), 1157. <https://doi.org/10.3390/bioengineering11111157>
7. Granvik, S. (2023). Open-source dog breed identification using CNN: explanation of the development & underlying technological specifications. [https://www.theseus.fi/bitstream/handle/10024/799064/Granvik\\_Samuel.pdf](https://www.theseus.fi/bitstream/handle/10024/799064/Granvik_Samuel.pdf)
8. Khan, N., Lee, M. Y., & Min, J. (2025). Head-Body Guided Deep Learning Framework for Dog Breed Recognition. *Computers, materials & continua*, 85(2), 2935-2958. <https://www.techscience.com/cmc/v85n2/63849/pdf>
9. Martins, O. O., Oosthuizen, C. C., & Desai, D. A. (2025). Evaluating unified training optimisations for MobileNetV2: Efficiency-accuracy trade-offs in fine-grained dog breed classification. *Discover Applied Sciences*, 7(11), 1240. <https://link.springer.com/article/10.1007/s42452-025-07798-1>
10. Valarmathi, B., Gupta, N. S., Prakash, G., Reddy, R. H., Saravanan, S., & Shanmugasundaram, P. (2023). Hybrid deep learning algorithms for dog breed identification—a comparative analysis. *IEEE Access*, 11, 77228-77239. <https://ieeexplore.ieee.org/document/10192536>
11. OpenAI. (2026). ChatGPT [Large language model]. <https://chat.openai.com/>
12. Google DeepMind. (2026). Gemini [Large language model]. <https://gemini.google.com/>
13. xAI. (2026). Grok [Large language model]. <https://grok.com/>
14. ml-debi. (2026). EfficientNetV2S-StanfordDogsA [Pretrained model]. Hugging Face. <https://huggingface.co/ml-debi/EfficientNetV2S-StanfordDogsA>